

STAT 6560 Final Project Supplementary Tables

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**Supplementary Figure 1: Scree plot from PCA on 7s data,
first 20 components**

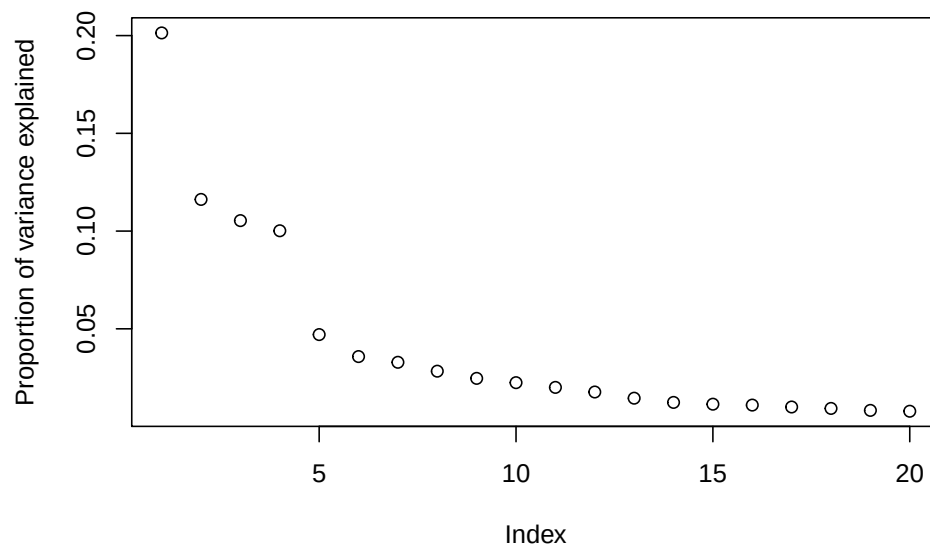


Table 1: Summary of first 20 components from 7s data

	Standard_deviation	Proportion_of_variance	Cumulative_proportion
Comp.1	3.6446964	0.2013615	0.2013615
Comp.2	2.7685524	0.1161875	0.3175490
Comp.3	2.6362477	0.1053480	0.4228970
Comp.4	2.5704248	0.1001529	0.5230499
Comp.5	1.7618918	0.0470557	0.5701056
Comp.6	1.5367765	0.0357994	0.6059050
Comp.7	1.4727501	0.0328785	0.6387835
Comp.8	1.3671655	0.0283332	0.6671167
Comp.9	1.2753257	0.0246545	0.6917712
Comp.10	1.2172737	0.0224611	0.7142323
Comp.11	1.1495649	0.0200318	0.7342641
Comp.12	1.0798753	0.0176767	0.7519408
Comp.13	0.9784205	0.0145113	0.7664521
Comp.14	0.9027697	0.0123540	0.7788061
Comp.15	0.8688472	0.0114430	0.7902491
Comp.16	0.8512258	0.0109836	0.8012326
Comp.17	0.8126532	0.0100107	0.8112433
Comp.18	0.7824241	0.0092798	0.8205231
Comp.19	0.7389042	0.0082762	0.8287993
Comp.20	0.7190925	0.0078383	0.8366376

**Supplementary Figure 2: Scree plot of PCA on 9s data,
first 20 components**

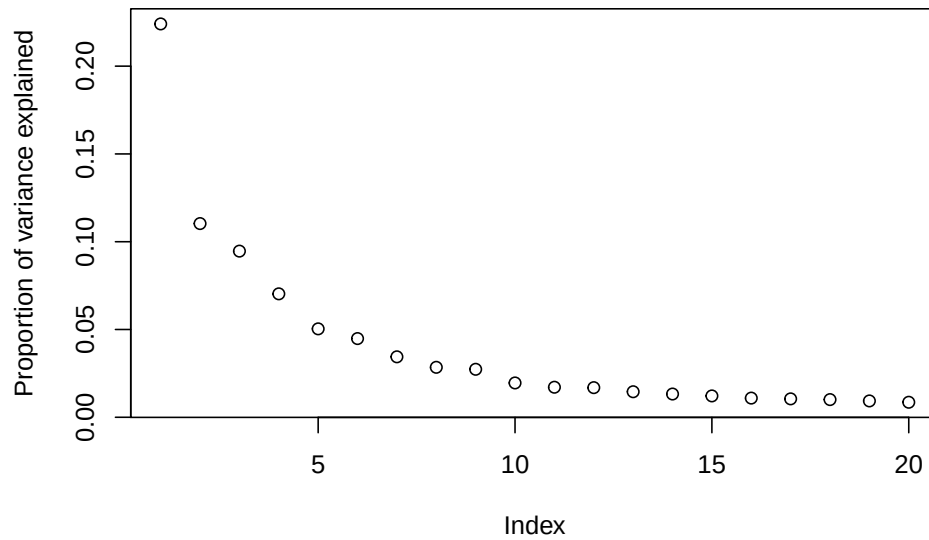


Table 2: Summary of first 20 components from 9s data

	Standard_deviation	Proportion_of_variance	Cumulative_proportion
Comp.1	3.9813048	0.2240492	0.2240492
Comp.2	2.7939026	0.1103354	0.3343846
Comp.3	2.5875255	0.0946372	0.4290218
Comp.4	2.2303179	0.0703114	0.4993332
Comp.5	1.8877068	0.0503688	0.5497020
Comp.6	1.7808957	0.0448301	0.5945321
Comp.7	1.5610263	0.0344439	0.6289761
Comp.8	1.4188436	0.0284552	0.6574312
Comp.9	1.3904874	0.0273292	0.6847604
Comp.10	1.1763019	0.0195583	0.7043187
Comp.11	1.1000995	0.0171063	0.7214250
Comp.12	1.0927349	0.0168780	0.7383030
Comp.13	1.0142729	0.0145413	0.7528443
Comp.14	0.9678893	0.0132417	0.7660860
Comp.15	0.9277911	0.0121673	0.7782533
Comp.16	0.8789713	0.0109205	0.7891738
Comp.17	0.8614233	0.0104888	0.7996626
Comp.18	0.8442136	0.0100739	0.8097364
Comp.19	0.8116387	0.0093115	0.8190479
Comp.20	0.7781982	0.0085600	0.8276079

Table 3: Confusion matrix for APER (LDA done on orig. variables)

	Predicted 7	Predicted 9
Actual 7	399	1
Actual 9	3	397

Table 4: Confusion matrix for LOO CV (LDA done on orig. variables)

	Predicted 7	Predicted 9
Actual 7	388	12
Actual 9	20	380

Table 5: Confusion matrix for testing (LDA done on orig. variables)

	Predicted 7	Predicted 9
Actual 7	243	157
Actual 9	2	87

Table 6: Confusion matrix for APER (LDA done on PCs)

	Predicted 7	Predicted 9
Actual 7	320	80
Actual 9	48	352

Table 7: Confusion matrix for LOOCV (LDA done on PCs)

	Predicted 7	Predicted 9
Actual 7	320	80
Actual 9	48	352

Table 8: Confusion matrix for testing (LDA done on PCs)

	Predicted 7	Predicted 9
Actual 7	209	191
Actual 9	14	75

Table 9: Summary of first 20 components from 7s and 9s data

	Standard_deviation	Proportion_of_variance	Cumulative_proportion
Comp.1	3.7330393	0.1846200	0.1846200
Comp.2	3.1239503	0.1292891	0.3139091
Comp.3	2.5690519	0.0874378	0.4013469
Comp.4	2.3142263	0.0709521	0.4722990
Comp.5	1.9579422	0.0507871	0.5230861
Comp.6	1.7766005	0.0418151	0.5649012
Comp.7	1.6340196	0.0353727	0.6002739
Comp.8	1.5327761	0.0311251	0.6313990
Comp.9	1.3811634	0.0252722	0.6566713
Comp.10	1.3435225	0.0239135	0.6805848
Comp.11	1.1733376	0.0182389	0.6988237
Comp.12	1.1141574	0.0164455	0.7152692
Comp.13	1.0688544	0.0151353	0.7304045
Comp.14	1.0536340	0.0147073	0.7451118
Comp.15	0.9686445	0.0124303	0.7575421
Comp.16	0.9315468	0.0114964	0.7690386
Comp.17	0.8903644	0.0105024	0.7795410
Comp.18	0.8528275	0.0096355	0.7891765
Comp.19	0.8419496	0.0093913	0.7985678
Comp.20	0.8244042	0.0090040	0.8075718